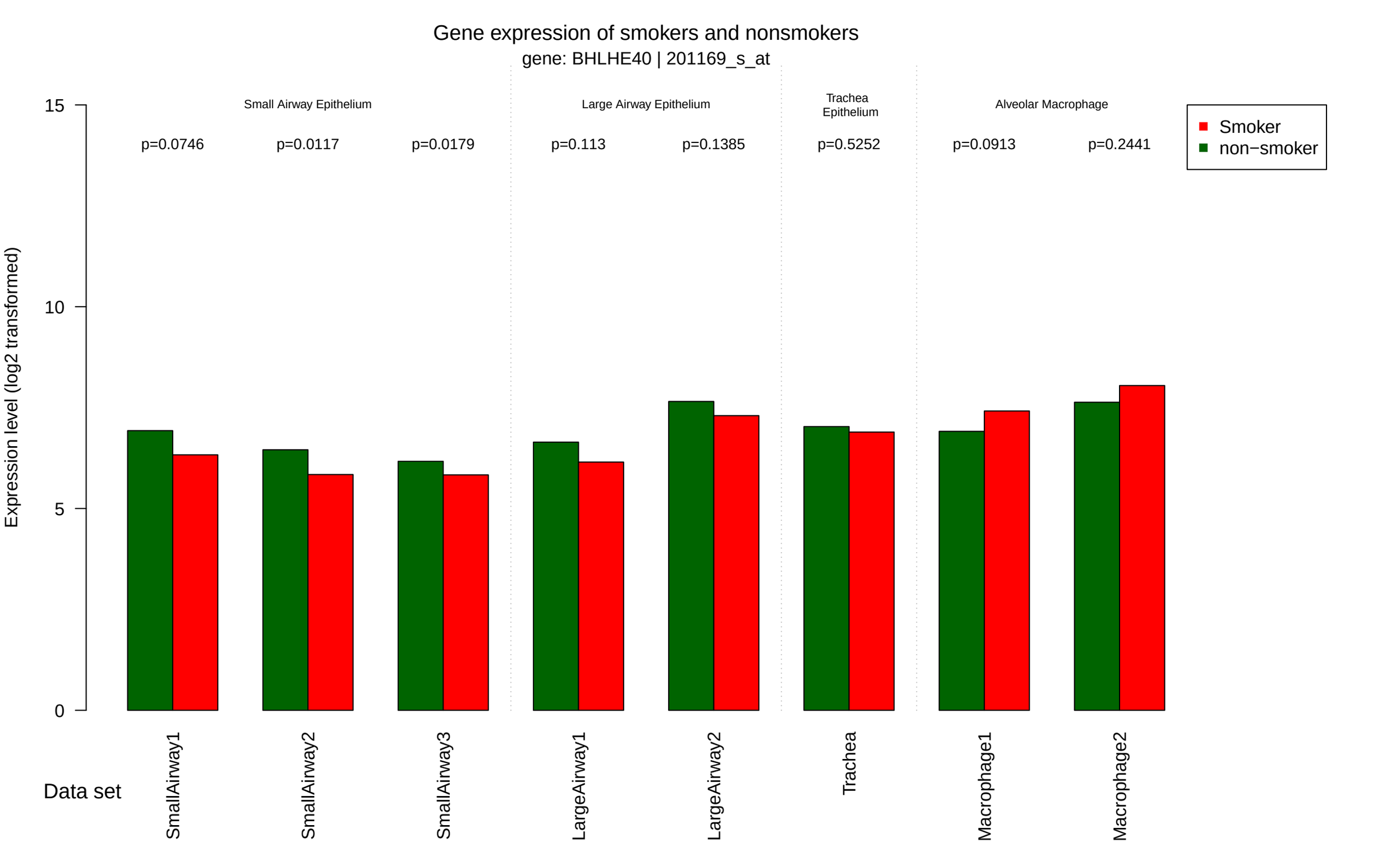
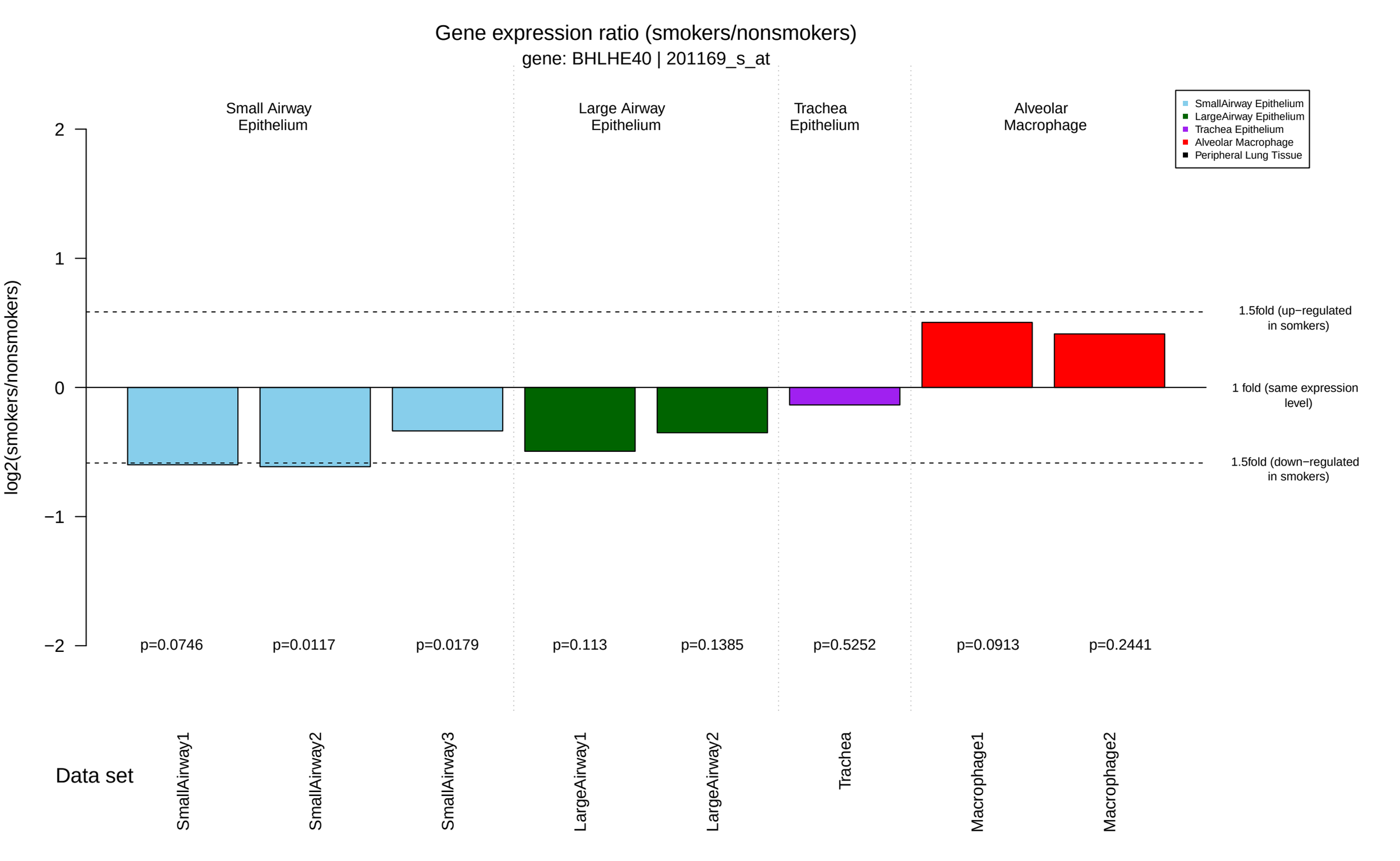
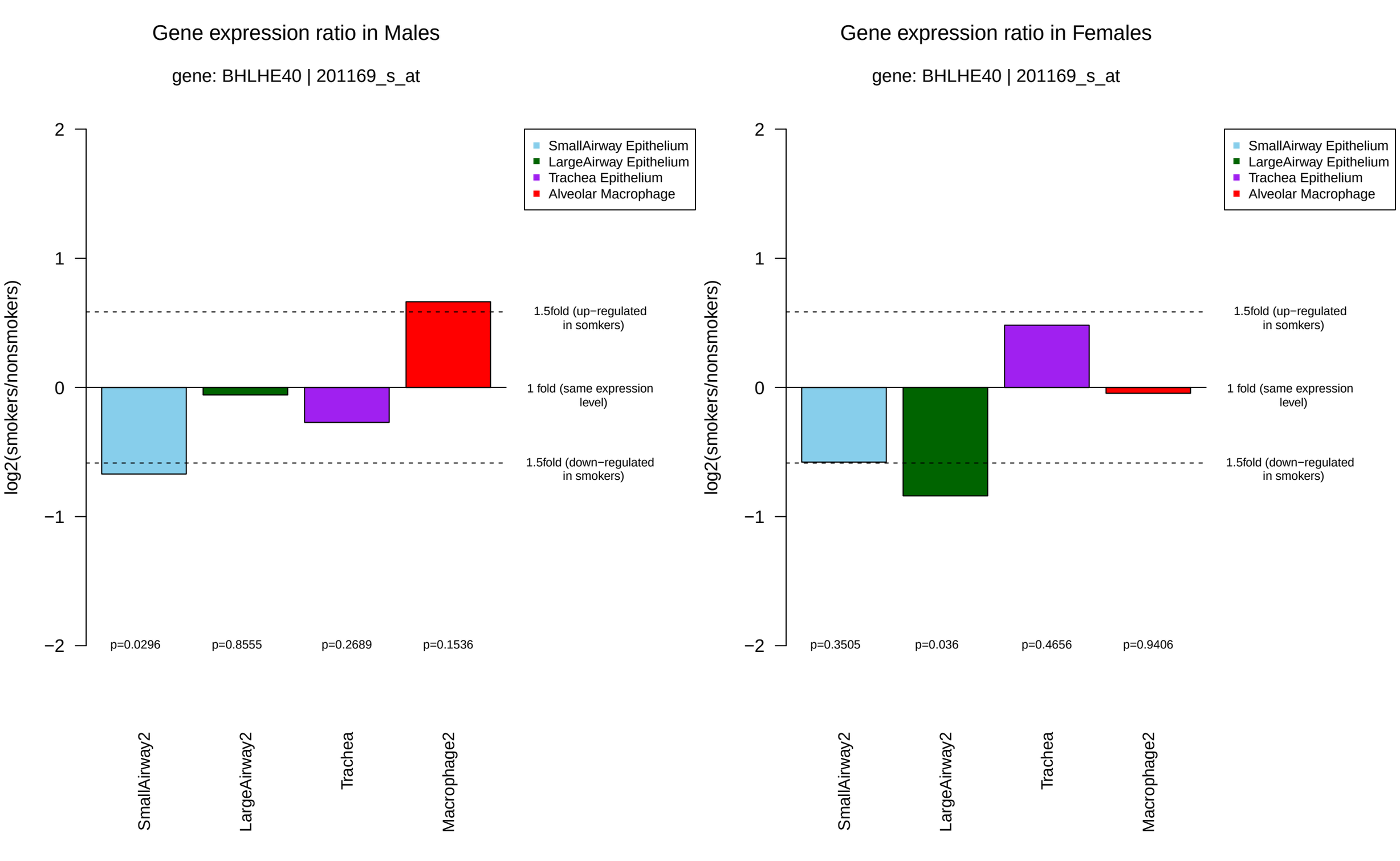
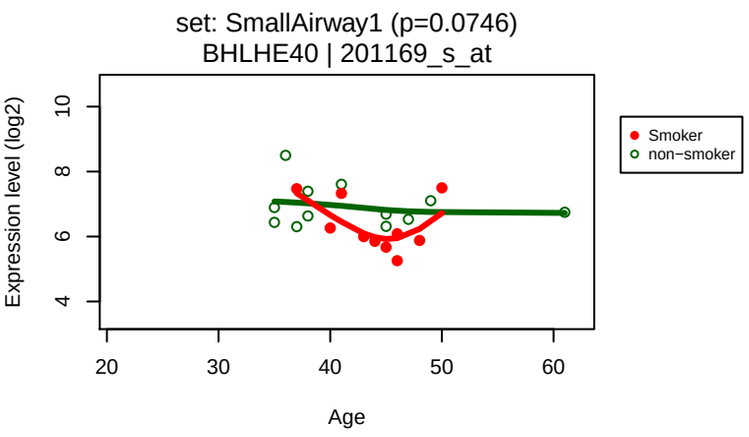
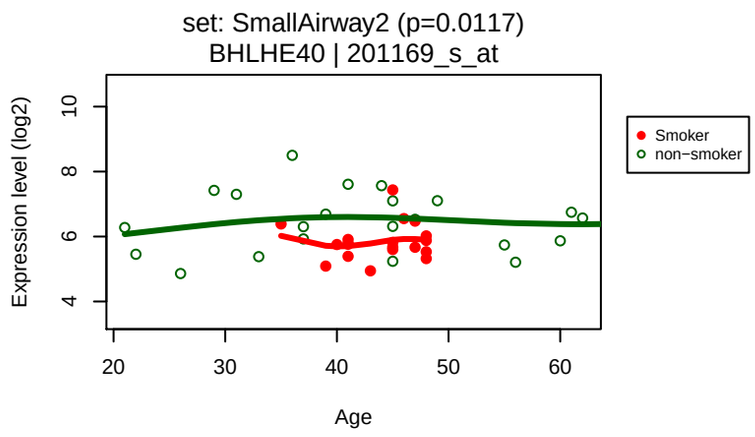
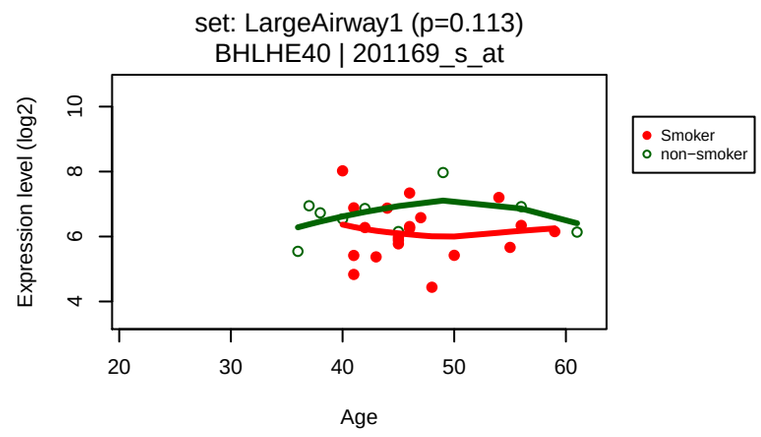

Gene expression of smokers and nonsmokers
gene: BHLHE40 | 201169_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0746
p=0.0117
p=0.0179
p=0.113
p=0.1385
p=0.5252
p=0.0913
p=0.2441
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

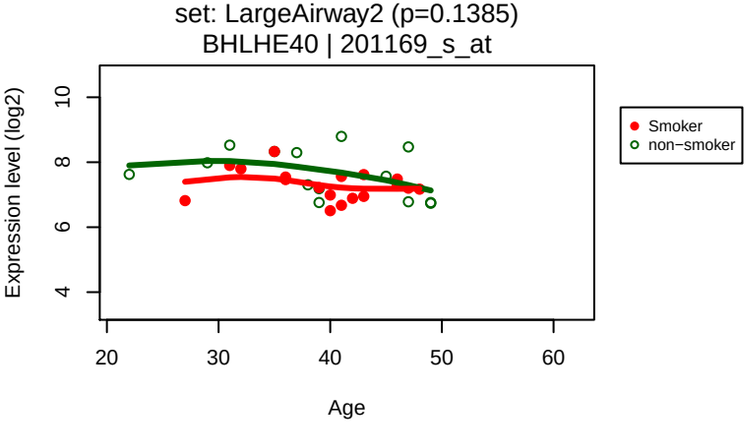

Gene expression ratio (smokers/nonsmokers)
gene: BHLHE40 | 201169_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0746
p=0.0117
p=0.0179
p=0.113
p=0.1385
p=0.5252
p=0.0913
p=0.2441
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

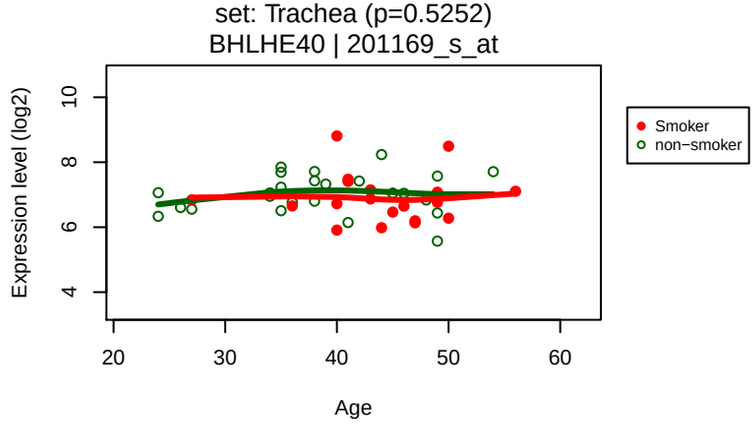

Gene expression ratio in Males
gene: BHLHE40 | 201169_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0296
p=0.8555
p=0.2689
p=0.1536
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BHLHE40 | 201169_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3505
p=0.036
p=0.4656
p=0.9406
SmallAirway2
LargeAirway2
Trachea
Macrophage2

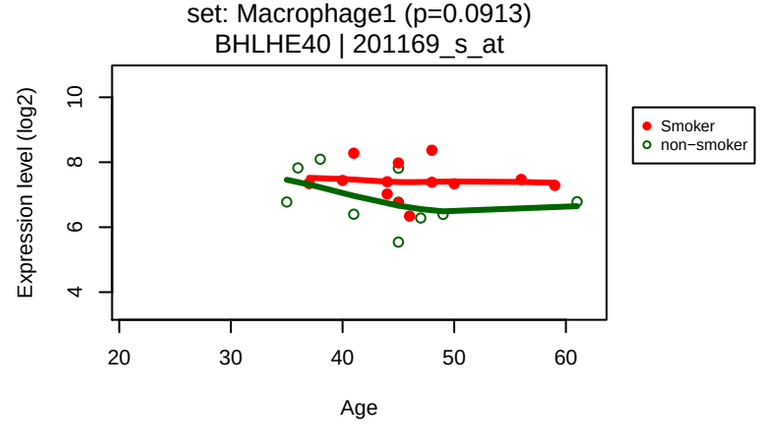

set: SmallAirway1 (p=0.0746)
BHLHE40 | 201169_s_at
Expression level (log2)
Age
Smoker
non−smoker

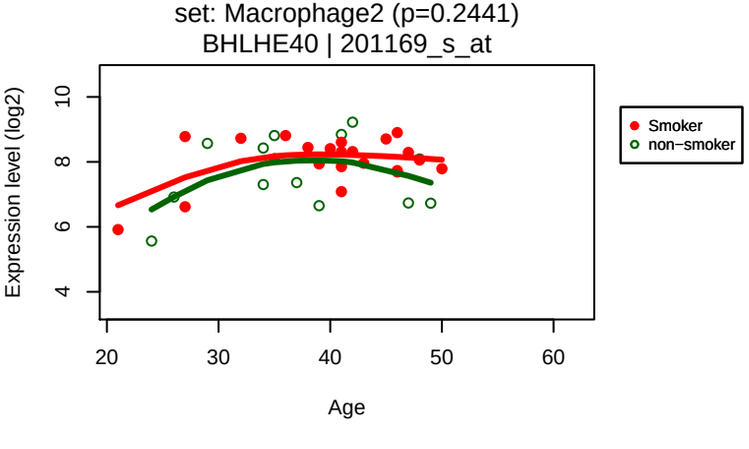

set: SmallAirway2 (p=0.0117)
BHLHE40 | 201169_s_at
Expression level (log2)
Age
Smoker
non−smoker

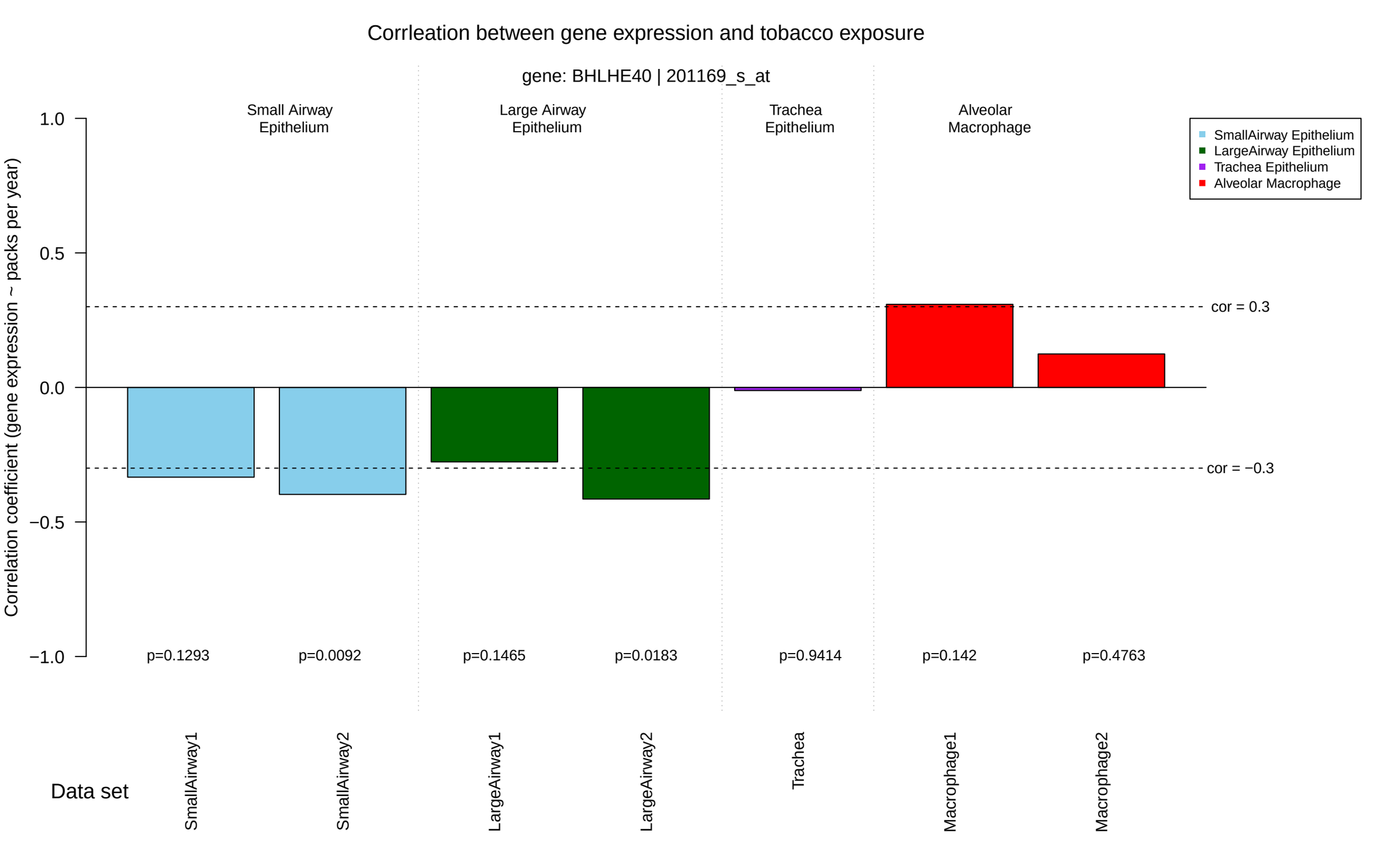

set: LargeAirway1 (p=0.113)
BHLHE40 | 201169_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1385)
BHLHE40 | 201169_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5252)
BHLHE40 | 201169_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0913)
BHLHE40 | 201169_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2441)
BHLHE40 | 201169_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BHLHE40 | 201169_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1293
p=0.0092
p=0.1465
p=0.0183
p=0.9414
p=0.142
p=0.4763
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2